

Tabular Foundation Models and Focused Semi-Supervised Learning under Label Scarcity

A controlled evaluation on ten OpenML classification datasets

960 new final-grid runs | balanced accuracy as the primary metric

Summary

This study tested whether modern tabular foundation models and targeted semi-supervised mechanisms improve prediction when only 50–500 labels are available. TabPFN-3 and TabICLv2 were evaluated in their standard few-shot form, then extended with self-training. Four trainable alternatives used the unlabeled training pool through graph regularization, retrieval attention, class-conditional representation alignment, or a combination of these mechanisms.

Main result. The two foundation models were the strongest and most reliable methods in the focused study. TabICLv2 achieved mean balanced accuracy 0.743 and TabPFN-3 0.739; their self-training variants reached 0.745 and 0.741. The gains from self-training were real but small and highly selective. Among the trainable SSL methods, unlabeled retrieval attention was best (0.668). Combining all regularizers did not improve on attention alone.

Against six established benchmark comparators—CatBoost, XGBoost, self-training LightGBM, VIME, SCARF, and SSLAE—a foundation-model variant was best in 31 of the 40 dataset-by-budget cells. The result is not that unlabeled data is uniformly beneficial: the strongest evidence is instead that frozen foundation models already provide a difficult baseline, and a conservative self-training guard can add small gains without broadly degrading it.

Experimental design

The benchmark used ten canonical OpenML datasets: phoneme, spambase, MagicTelescope, adult, bank-marketing, electricity, satimage, segment, steel-plates-fault, and jannis. Every method used the same predetermined split for each dataset, seed, and label budget. The grid contained four budgets (50, 100, 250, 500) and three seeds. Phase A evaluated two frozen foundation models (240 runs); Phase C evaluated six SSL methods (720 runs). All 960 runs completed successfully, with no missing or duplicate keys.

The protocol was inductive. Graphs, pseudo-labels, representation learning, and retrieval memory used labeled and unlabeled *training* rows only. Validation labels were restricted to calibration, early stopping, and safe fallback decisions. Test rows were introduced only as prediction queries. TabPFN-3 and TabICLv2 used the native mixed-type pandas representation; the trainable methods used the benchmark’s established imputed, scaled, and one-hot encoded representation. Accuracy, macro-F1, log-loss, ROC-AUC, average precision where defined, Brier score, ECE, and runtime were retained as secondary outcomes.

What was implemented

Method	Implementation used in the final grid
tabpfn3	TabPFN 8.1.0 using the verified <code>tabpfn-v3-classifier-v3_default.ckpt</code> ; labeled context only, native feature view, ordered and validated class probabilities.
tabicl v2	Pinned <code>tabicl-classifier-v2-20260212.ckpt</code> ; labeled context only and the same native-view, few-shot protocol.
tabpfn3_self_training and tabicl v2_self_training	One shared, deterministic engine with up to three hard pseudo-label rounds, class-balanced confidence selection, per-class and total caps, validation-based round selection, and fallback to the frozen predictor.
laplacian_ssl	Trainable encoder/classifier with a sparse training-only kNN graph, locally scaled affinities, mini-batched Laplacian loss, ramp-up, validation stopping, and collapse checks.
unlabeled_attention_ssl	Trainable encoder with batched top- <i>k</i> retrieval and residual attention over labeled and unlabeled training memory; self-neighbors and validation/test rows were excluded.
embedding_alignment_ssl	Confidence-weighted, class-conditional prototype attraction with inter-class separation and representation-collapse diagnostics.
geometric_attention_ssl	Modular combination of supervised loss, pseudo-label consistency, Laplacian regularization, retrieval attention, prototype alignment, and class separation, with confidence and loss ramp-up.

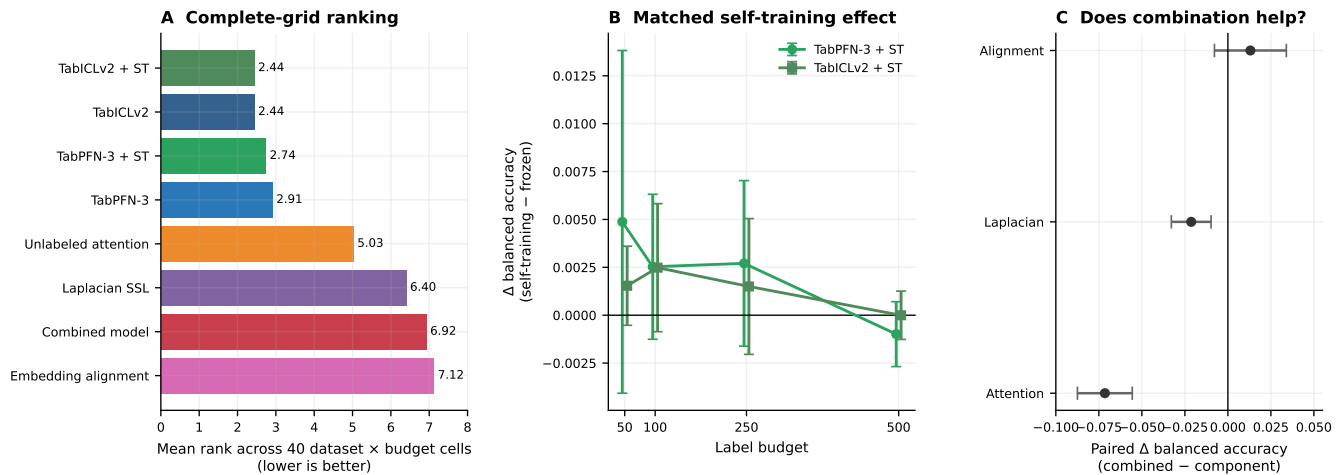
Results

Complete-grid results for the eight requested methods

Method	Bal. acc.	Mean rank	Macro-F1	ECE	Time/run (s)
TabICLv2 + self-training	0.745	2.44	0.747	0.045	11.1
TabICLv2	0.743	2.44	0.745	0.044	6.3
TabPFN-3 + self-training	0.741	2.74	0.743	0.040	13.3
TabPFN-3	0.739	2.91	0.741	0.039	8.8
Unlabeled attention	0.668	5.03	0.657	0.122	5.0
Laplacian SSL	0.617	6.40	0.602	0.099	7.6
Combined model	0.596	6.92	0.569	0.082	9.5
Embedding alignment	0.583	7.12	0.559	0.111	3.6

Means pool 120 successful runs per method. Mean rank first averages the three seeds within each of the 40 dataset-by-budget cells and then ranks methods within each cell; lower is better. ECE is expected calibration error.

Foundation-model ranking and the incremental value of unlabeled-data mechanisms



A: failure-aware ranking over the requested methods. **B:** exact paired self-training effects; bars are descriptive 95% intervals across matched dataset/seed cells. **C:** the combined model compared with separately trained components on identical run keys.

Foundation models and self-training

TabICLv2 was slightly stronger than TabPFN-3: the exact paired difference TabPFN-3 minus TabICLv2 was -0.0048 balanced accuracy (descriptive 95% interval $-0.0092, -0.0003$). Self-training increased mean balanced accuracy by $+0.0014$ for TabICLv2 and $+0.0023$ for TabPFN-3. These averages hide how conservative the mechanism was. The selected prediction was identical to the frozen model in 83/120 TabICLv2 cells and 86/120 TabPFN-3 cells; positive changes occurred in only 23 and 20 cells, respectively. The largest consistent benefit occurred on steel-plates-fault, with smaller gains on bank-marketing. At 500 labels, neither backbone showed a mean benefit.

The foundation-model family won 31/40 cells when compared with the six key historical baselines. Pooled balanced accuracy was 0.722 for CatBoost, 0.712 for XGBoost, 0.690 for SSLAE, 0.687 for SCARF, 0.686 for VIME, and 0.675 for self-training LightGBM. These are benchmark-level comparisons rather than claims of statistical dominance on every dataset.

Trainable SSL mechanisms

Unlabeled attention was the strongest trainable SSL method and won three cells within the focused eight-method grid, including adult at budgets 100 and 250. Its mean attention mass was 96% on unlabeled memory, confirming that the final registration genuinely used the unlabeled pool. Laplacian SSL was modestly better than historical label propagation ($+0.014$, interval crossing zero) but worse than label spreading (-0.064). Embedding alignment was not competitive as a standalone predictor.

The full combined model improved on embedding alignment by $+0.013$ on average, but underperformed Laplacian SSL by -0.021 and unlabeled attention by -0.072 . More regularization therefore did not translate into better generalization.

Reliability, diagnostics, and interpretation

Repair of the combined model

An early segment run collapsed to exactly chance performance (1/7 balanced accuracy). The failure was reproduced before Phase C and traced to retrieval composition: because the unlabeled pool was much larger, an unrestricted top- k search could fill every memory slot with unlabeled examples. Class mapping, output dimension, and probability ordering were verified separately. The repair introduced deterministic balanced retrieval from labeled and unlabeled memory, batched full-pool diagnostics, supervised warm-up, and confidence/loss ramp-up. A final 12-cell segment/Jannis gate passed before the full attention-family grid was submitted.

Across the final grid, no method triggered a representation-collapse flag and no attention memory contained validation or test rows. The combined model maintained effective embedding rank 32 and allocated mean attention mass 55% to labeled and 45% to unlabeled neighbors. It nevertheless triggered 21/120 constant-prediction warnings; Laplacian SSL triggered 17/120. These warnings were retained rather than filtered because their probabilities were finite and the runs satisfied the prespecified operational checks. They help explain why stable optimization did not guarantee competitive accuracy.

The sparse graphs contained about 21,000 training nodes and 53,000 edges on average and took 1.7 seconds to construct. The foundation models were also the best calibrated methods (mean ECE 0.039–0.045), while trainable SSL ECE ranged from 0.082 to 0.122. Average runtime per final run ranged from 3.6 seconds for alignment to 13.3 seconds for TabPFN-3 self-training. For context, historical SSLAE, SCARF, and VIME runs averaged 57, 177, and 263 seconds.

What can and cannot be concluded

- **Supported:** under this label-budget protocol, TabPFN-3 and TabICLv2 are stronger overall than the focused trainable SSL implementations and the selected historical baselines.
- **Supported:** validation-guarded TFM self-training is safe and can produce small gains, but most cells revert to the frozen model and the benefit largely disappears at 500 labels.
- **Supported:** retrieval attention is the most promising trainable unlabeled-data mechanism in this study; combining all mechanisms is not beneficial.
- **Not supported:** a general claim that unlabeled data helps every method. The trainable SSL grid did not include a final-source-hash supervised encoder control for every architecture, so its results are benchmark comparisons rather than pure causal estimates of unlabeled-data value.

Three seeds permit matched descriptive uncertainty but not strong significance claims for small differences. CPU RAM and reliable peak-GPU-memory summaries were not serialized. Post-training oracle pseudo-label accuracy, neighbor-label purity, and class-prior drift were also not consistently available as analysis fields. Average precision is undefined for the multiclass tasks and was left missing rather than imputed.

Conclusion

The practical result is straightforward: use a frozen tabular foundation model as the primary low-label baseline, with conservative self-training as an optional refinement. TabICLv2 was marginally strongest overall; TabPFN-3 was very close and slightly better calibrated. If a trainable SSL alternative is needed, retrieval attention is the best candidate from this study. The Laplacian, prototype-alignment, and complete combined variants add complexity without improving the final ranking.

Reproducibility

All 240 Phase-A and 720 Phase-C shards were validated before aggregation. The final test suite passed 25 tests with one skipped. Canonical results, machine-readable paired effects, figures, source/configuration hashes, checkpoint identities, Slurm metadata, and the full technical report are preserved under `results/reports/bar_ilan_tfm_ssl_final/`. The historical 17-method result file was not modified.